

AIM:

Binary Search with Big-O Analysis

main.c

Run ▶

```
1  #include <stdio.h>
2
3  int main() {
4      int a[50], n, key;
5      int low, high, mid;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter sorted elements:\n");
11     for(int i = 0; i < n; i++)
12         scanf("%d", &a[i]);
13
14     printf("Enter element to search: ");
15     scanf("%d", &key);
16
17     low = 0;
18     high = n - 1;
19
20     while(low <= high) {
21         mid = (low + high) / 2;
```

Output

```
Enter number of elements: 6
Enter sorted elements:
23 34 45 56 67 78
Enter element to search: 56
Element found at position 4
```

Result:

Binary Search with Big-O Analysis has been executed successfully